JUUM

STRUMPLATE SYNTHESIZER · USER MANUAL

PLAY — Tap or drag across the blue strumplate to trigger notes.

SHAPE — Tune two voices, filters, envelopes, and experimental tone controls.

EXPLORE — Choose chords, use the arpeggiator, or build a 16-step pattern.

CONNECT — Explore MIDI features where your browser and devices support them.

JUUM is a browser-based synthesizer designed for hands-on sound exploration. This guide reflects the current online version and its visible controls.

Online version: https://mentlone.github.io/JUUM/

Revision: October 22, 2026 • Use a current browser. Feature availability can vary by device.

1. Quick Start

Make your first sound

• Open JUUM in a modern browser such as Chrome, Edge, or Safari.

• Raise your device volume and check which audio output is selected.

• Tap or click the large strumplate. Drag across it to play as you move.

• Press **TEST TONE** to check audio; it should play a short A4 note.

• Press **MAKE NOISE** for a brief noise burst to check the output path.

Browsers generally require a user gesture before audio can start. If JUUM is silent, interact with the page once, confirm volume and output, and reload if needed.

Playing the strumplate

The strumplate is a touch/mouse playing surface. Tap or click to trigger notes; dragging across the surface creates a continuous performance gesture. It is a software control, not a physical sensor.

2. Voices & Tone Controls

Main Voice

The primary oscillator provides the main tone. Use its waveform selector and controls for volume, cutoff, resonance, attack, decay, sustain, and release. The Main Voice switch enables or mutes this layer.

Sub Voice

The Sub Voice adds a lower oscillator layer, normally one octave below the played note. Its waveform, volume, filter, and envelope let you create bass weight or a blended second tone. Disable it to hear the Main Voice alone.

Wave Character

Wave Character is an experimental tone-shaping control that alters the character of the main oscillator. Sweep it slowly while holding or retriggering notes to hear the change; the effect depends on the selected waveform and settings.

Detune

Detune shifts the Sub Voice by cents relative to its normal pitch. Small positive or negative values create a thicker, beating tone; larger shifts sound more obviously out of tune.

Knob operation

Drag a knob vertically with a mouse or finger to change its value. Use the displayed readout as a reference. If a sound becomes too extreme, return the relevant knob toward its starting value.

3. Global, Chords & Arpeggiator

Root note and scale

Use the Global controls to choose the root note and scale that inform the chord and note material. The chord buttons select the current harmony for playing and sequencing.

Arpeggiator

• **Rate:** controls how quickly notes are played.

• **Octaves:** sets the span of the repeating pattern.

• **Swing:** adds rhythmic lilt.

• **Gate:** controls how long each arpeggiated note is held.

• **Mode:** Up, Down, Up/Down, or Random.

Use the ARP control to start or stop the pattern. Try changing the chord, rate, octave span, and mode while listening.

4. Effects & Output

Reverb

Adds a sense of space and ambience. Turn it up gradually to keep notes clear while adding depth.

Delay

Repeats the sound. Delay Time changes the spacing between repeats. Longer settings can create rhythmic trails.

Phase Bloom

An experimental, moving resonant shimmer effect. Raise its amount gradually and compare the sound with the effect switch enabled and disabled.

FX Volume & Master

FX Volume controls the overall effects return level. Master controls the overall synth output. Effect switches bypass their effects while leaving knob settings available. An effect may sound inactive if its amount or return level is near zero.

5. Presets & Waveform Display

Preset Memory

Use Save Preset to name and store the current sound. Select a saved preset, then use Load or Delete as needed. Presets are stored in this browser's local storage and do not automatically sync to other browsers or devices.

Live Waveform

The waveform display visualizes the audio output. Play a note or press TEST TONE to activate the audio engine and help the display respond.

6. 16-Step Sequencer

Building a pattern

• Choose a chord and set the preferred Root note and Scale in Global.

• Click numbered steps 01–16 to toggle them on or off. Enabled steps light up.

• For a simple four-on-the-floor pattern, try steps 01, 05, 09, and 13.

• Set the sequencer BPM, press PLAY, and press STOP when finished.

The sequencer currently generates notes automatically from the selected chord; individual pitches cannot yet be assigned to each step. If you hear no sound, enable at least one step, select a chord, make sure a voice is enabled, and check voice and Master volume.

7. MIDI & Clock

MIDI behavior depends on browser, operating system, permissions, and connected equipment. MIDI access generally requires a compatible browser/device and a secure HTTPS page.

CONTROL	WHAT IT DOES
MIDI	Requests MIDI access and checks for available outputs.
Send MIDI clock	Enables outgoing MIDI clock messages.
Play / Stop	Starts or stops outgoing MIDI clock; it does not start the internal audio synthesizer.
MIDI Thru	Passes supported incoming notes to the selected output.
Receive clock	Enables incoming clock detection where supported.
Send notes	Enables note messages to the MIDI output.

8. Troubleshooting

No sound

• Click TEST TONE once to unlock browser audio.

• Raise device volume and confirm the correct output is selected.

• Check Master and Main Voice Volume; make sure at least one voice is enabled.

• For the sequencer, enable one or more steps and select a chord.

• Try a current version of Chrome, Edge, or Safari, then reload the page.

Controls are hard to use on a phone

Tap the strumplate and drag knobs vertically. Rotate the device if that gives you a more comfortable layout, and keep the browser tab active while playing.

What is experimental?

JUUM is an evolving project. Browser audio and MIDI behavior varies by device. The strumplate is a touch/mouse interface, and features may change as development continues.

Online version: https://mentlone.github.io/JUUM/
Source code: github.com/MENTLONE/JUUM